

Seminar 1. Bayesian Reasoning: Solutions

Bayesian Methods

1. A medical test finds a serious disease in a patient. The test is 99% accurate: if the patient has the disease, the test is positive with probability 99%; if the patient does not have the disease, the test is negative with probability 99%. The disease is rare: only one person in 10 000 has it. Find the probability that the patient really has the disease.

Interpretation. Let $D \in \{0, 1\}$ show whether the patient has the disease ($D = 1$). Let $T \in \{0, 1\}$ be the test result ($T = 1$ is a positive result). The problem gives three values:

$$p(D = 1) = 10^{-4}, \quad p(T = 1 \mid D = 1) = 0.99, \quad p(T = 0 \mid D = 0) = 0.99.$$

Find $p(D = 1 \mid T = 1)$.

Idea. The disease causes the test result, so the causal graph is $D \rightarrow T$. The graph gives the joint distribution as a product of two factors:

$$p(D, T) = p(D) p(T \mid D).$$

The problem gives both factors: the distribution of the source node D and the conditional distribution on the edge $D \rightarrow T$. Thus the joint distribution is known. From the joint distribution, you can get any other probability with two operations:

- A marginal probability is the sum of the joint probability over the other variables:

$$p(T) = \sum_D p(D, T).$$

For continuous variables x and y , the sum is an integral: $p(y) = \int p(x, y) dx$.

- A conditional probability is a joint probability divided by a marginal probability:

$$p(D \mid T) = \frac{p(D, T)}{p(T)}.$$

These two operations together give Bayes' theorem:

$$p(D \mid T) = \frac{p(D, T)}{p(T)} = \frac{p(D, T)}{\sum_{D'} p(D', T)} = \frac{p(D) p(T \mid D)}{\sum_{D'} p(D') p(T \mid D')}.$$

Solution.

Step 1. Find the missing probabilities.

The probabilities of the two values of a variable add to 1. Thus

$$\begin{aligned} p(D = 0) &= 1 - p(D = 1) = 1 - 0.0001 = 0.9999, \\ p(T = 1 \mid D = 0) &= 1 - p(T = 0 \mid D = 0) = 1 - 0.99 = 0.01. \end{aligned}$$

The second value is the probability of a false positive result.

Step 2. Find the joint probabilities of a positive result.

Use $p(D, T) = p(D) p(T \mid D)$:

$$\begin{aligned} p(D = 1, T = 1) &= p(D = 1) p(T = 1 \mid D = 1) = 0.0001 \cdot 0.99 = 0.000099, \\ p(D = 0, T = 1) &= p(D = 0) p(T = 1 \mid D = 0) = 0.9999 \cdot 0.01 = 0.009999. \end{aligned}$$

Step 3. Find the marginal probability of a positive result.

Sum the joint probabilities over D :

$$p(T = 1) = p(D = 1, T = 1) + p(D = 0, T = 1) = 0.000099 + 0.009999 = 0.010098.$$

Step 4. Divide the joint probability by the marginal probability.

$$p(D = 1 \mid T = 1) = \frac{p(D = 1, T = 1)}{p(T = 1)} = \frac{0.000099}{0.010098} = \frac{99}{10098} = \frac{99}{99 \cdot 102} = \frac{1}{102} \approx 0.0098.$$

The patient has the disease with a probability of approximately 1%.

Conclusion. A positive result increases the probability of the disease from 0.01% to approximately 1%. But the patient is still healthy with a probability of 99%. The cause is the low prior probability. In 10 000 people, the test finds approximately $10\,000 \cdot 0.000099 \approx 1$ sick person and $10\,000 \cdot 0.009999 \approx 100$ healthy people. For this reason, most positive results of a mass test for a rare disease are false.

A second independent test gives more information. After the first positive result, the probability of the disease is $1/102$. This value is the prior probability for the second test. If the second result is also positive, then

$$p(D = 1 \mid T_1 = 1, T_2 = 1) = \frac{\frac{1}{102} \cdot 0.99}{\frac{1}{102} \cdot 0.99 + \frac{101}{102} \cdot 0.01} = \frac{0.99}{0.99 + 1.01} = 0.495.$$

2. Consider the following probabilistic model. At a seminar, a student answers badly ($A = 1$) or well ($A = 0$). This depends on whether the student has depression ($D = 1$ or $D = 0$) and whether the student went to a party the night before ($P = 1$ or $P = 0$). The party can also give the student a headache ($H = 1$). A bad answer makes the teacher unhappy ($T = 1$). Figure 1 shows the causal relations in the model. The probabilities are:

$p(A = 1 \mid P, D) \mid P \quad D$	$p(H = 1 \mid P) \mid P$	$p(T = 1 \mid A) \mid A$
0.999	0.9	0.95
0.9	0.2	0.5
0.9		
0.01		

$$p(P = 1) = 0.2, \quad p(D = 1) = 0.4.$$

Find $p(P = 1 \mid H = 1)$, $p(P = 1 \mid T = 1)$, and $p(P = 1 \mid T = 1, H = 1)$.

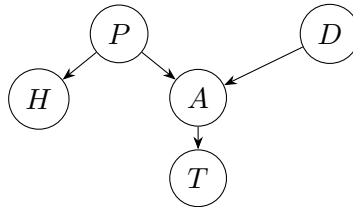


Figure 1: Causal relations for Problem 2

Idea. This problem uses the method of Problem 1 on a larger graph. First, write the joint distribution with the graph in Figure 1.

The chain rule is true for any distribution and any order of the variables. Use an order where each parent comes before its children:

$$p(P, D, A, H, T) = p(P) p(D | P) p(A | P, D) p(H | P, D, A) p(T | P, D, A, H).$$

The graph adds the assumptions of the model. An arrow shows a direct influence. If there is no arrow, there is no direct influence. Thus, given its parents, a variable does not depend on the variables before it in the order. Apply this rule to each factor:

- D has no parents, so $p(D | P) = p(D)$. The depression does not depend on the party.
- The parents of A are P and D , so the factor $p(A | P, D)$ does not change.
- The only parent of H is P , so $p(H | P, D, A) = p(H | P)$. Only the party causes the headache.
- The only parent of T is A , so $p(T | P, D, A, H) = p(T | A)$. The teacher sees only the answer.

Substitute these factors into the chain rule:

$$p(P, D, A, H, T) = p(P) p(D) p(A | P, D) p(H | P) p(T | A).$$

This is the general rule for a directed graph without cycles: the joint distribution is the product of the factors $p(\text{node} | \text{parents})$. A model of this type is a *Bayesian network*.

The order is important. In a different order, a factor can have a child in the condition, for example $p(A | T)$. The graph does not simplify this factor, and the tables do not give it. In the order from parents to children, each factor becomes one of the given tables.

The tables give all five factors. Thus the joint distribution is known, and you can get any probability from it. To find the probability of P given an observation, move through the graph in two directions:

- Along the edges, from P to the observed variable. Sum over each unobserved variable on the path, for example $p(T | P) = \sum_A p(T | A) p(A | P)$.
- Against the edges, from the observed variable back to P . Use Bayes' theorem, for example $p(P | T) = p(P) p(T | P) / p(T)$.

Each factor $p(\text{node} | \text{parents})$ sums to 1 over the values of the node. Thus, when you sum the joint distribution over a variable that is not a parent of other variables, its factor disappears.

Solution.

Step 1. Find the missing probabilities.

The probabilities of the two values of a variable add to 1:

$$p(P = 0) = 1 - 0.2 = 0.8, \quad p(D = 0) = 1 - 0.4 = 0.6.$$

Step 2. Show that P and D are independent.

Sum the joint distribution over T , then H , then A . Each sum removes one factor, because the factor sums to 1:

$$p(P, D) = \sum_A \sum_H \sum_T p(P, D, A, H, T) = p(P) p(D).$$

Thus $p(D | P) = p(P, D) / p(P) = p(D)$.

Step 3. Write $p(A | P)$ as a sum over D .

The depression D is not observed. Sum over D , then use the definition of a conditional probability, then use Step 2:

$$p(A | P) = \sum_D p(A, D | P) = \sum_D p(A | P, D) p(D | P) = \sum_D p(A | P, D) p(D).$$

Step 4. Calculate $p(A = 1 | P)$.

Substitute the values from the tables and from Step 1:

$$\begin{aligned} p(A = 1 | P = 1) &= 0.999 \cdot 0.4 + 0.9 \cdot 0.6 = 0.3996 + 0.54 = 0.9396, \\ p(A = 1 | P = 0) &= 0.9 \cdot 0.4 + 0.01 \cdot 0.6 = 0.36 + 0.006 = 0.366. \end{aligned}$$

Thus $p(A = 0 | P = 1) = 1 - 0.9396 = 0.0604$ and $p(A = 0 | P = 0) = 1 - 0.366 = 0.634$.

Step 5. Show that $p(T | A, P) = p(T | A)$.

Sum the joint distribution over H and D . Then do the same without T :

$$\begin{aligned} p(P, A, T) &= p(P) p(T | A) \sum_D p(D) p(A | P, D), \\ p(P, A) &= p(P) \sum_D p(D) p(A | P, D). \end{aligned}$$

Divide the first line by the second line: $p(T | A, P) = p(P, A, T)/p(P, A) = p(T | A)$.

Step 6. Write $p(T | P)$ as a sum over A .

The answer A is not observed. Sum over A , then use the definition of a conditional probability, then use Step 5:

$$p(T | P) = \sum_A p(T, A | P) = \sum_A p(T | A, P) p(A | P) = \sum_A p(T | A) p(A | P).$$

Step 7. Calculate $p(T = 1 | P)$.

Substitute the values from the table and from Step 4:

$$\begin{aligned} p(T = 1 | P = 1) &= 0.95 \cdot 0.9396 + 0.5 \cdot 0.0604 = 0.89262 + 0.0302 = 0.92282, \\ p(T = 1 | P = 0) &= 0.95 \cdot 0.366 + 0.5 \cdot 0.634 = 0.3477 + 0.317 = 0.6647. \end{aligned}$$

Step 8. Find $p(P = 1 | H = 1)$.

Sum the joint distribution over T , A , and D . Each sum removes one factor, so $p(P, H) = p(P) p(H | P)$. Calculate the joint probabilities:

$$\begin{aligned} p(P = 1, H = 1) &= 0.2 \cdot 0.9 = 0.18, \\ p(P = 0, H = 1) &= 0.8 \cdot 0.2 = 0.16. \end{aligned}$$

Sum them over P : $p(H = 1) = 0.18 + 0.16 = 0.34$. Divide:

$$p(P = 1 | H = 1) = \frac{p(P = 1, H = 1)}{p(H = 1)} = \frac{0.18}{0.34} = \frac{9}{17} \approx 0.529.$$

Step 9. Find $p(P = 1 | T = 1)$.

Use $p(P, T) = p(P) p(T | P)$ and the values from Step 7:

$$\begin{aligned} p(P = 1, T = 1) &= 0.2 \cdot 0.92282 = 0.184564, \\ p(P = 0, T = 1) &= 0.8 \cdot 0.6647 = 0.53176. \end{aligned}$$

Sum them over P : $p(T = 1) = 0.184564 + 0.53176 = 0.716324$. Divide:

$$p(P = 1 | T = 1) = \frac{p(P = 1, T = 1)}{p(T = 1)} = \frac{0.184564}{0.716324} \approx 0.258.$$

Step 10. Show that H and T are independent given P .

Sum the joint distribution over the unobserved variables D and A :

$$\begin{aligned}
 p(P, H, T) &= \sum_A \sum_D p(P) p(D) p(A | P, D) p(H | P) p(T | A) \\
 &= p(P) p(H | P) \sum_A p(T | A) \sum_D p(D) p(A | P, D) \quad (\text{move factors out of the sums}) \\
 &= p(P) p(H | P) \sum_A p(T | A) p(A | P) \quad (\text{Step 3}) \\
 &= p(P) p(H | P) p(T | P) \quad (\text{Step 6}).
 \end{aligned}$$

Divide by $p(P)$: $p(H, T | P) = p(H | P) p(T | P)$.

Step 11. Find $p(P = 1 | T = 1, H = 1)$.

Use the product from Step 10 and the values from Step 7:

$$\begin{aligned}
 p(P = 1, H = 1, T = 1) &= 0.2 \cdot 0.9 \cdot 0.92282 = 0.166108, \\
 p(P = 0, H = 1, T = 1) &= 0.8 \cdot 0.2 \cdot 0.6647 = 0.106352.
 \end{aligned}$$

Sum them over P : $p(H = 1, T = 1) = 0.166108 + 0.106352 = 0.27246$. Divide:

$$p(P = 1 | T = 1, H = 1) = \frac{p(P = 1, H = 1, T = 1)}{p(H = 1, T = 1)} = \frac{0.166108}{0.27246} \approx 0.610.$$

Conclusion. The prior probability of the party is 0.2. A headache increases it to 0.529. An unhappy teacher increases it only to 0.258, because a bad answer has other causes, and the teacher is often unhappy without a bad answer. The two observations together give the highest probability, 0.610.

3. Let $x_1 \sim \text{Poiss}(\lambda_1)$ and $x_2 \sim \text{Poiss}(\lambda_2)$ be two independent Poisson random variables, that is, $p(x_i = k) = \exp(-\lambda_i) \frac{\lambda_i^k}{k!}$, $k = 0, 1, 2, \dots$. Prove that $x_1 + x_2 \sim \text{Poiss}(\lambda_1 + \lambda_2)$.

Idea. To find the distribution of a function $y = f(x_1, \dots, x_n)$ of discrete random variables, find $p(y = k)$ for each value k :

- Write the event $\{f(x_1, \dots, x_n) = k\}$ as a union of disjoint events $\{x_1 = j_1, \dots, x_n = j_n\}$, one for each tuple with $f(j_1, \dots, j_n) = k$.
- The probability of a union of disjoint events is the sum of their probabilities:

$$p(y = k) = \sum_{f(j_1, \dots, j_n) = k} p(x_1 = j_1, \dots, x_n = j_n).$$

- If the variables are independent, write each joint probability as a product.
- Simplify the sum and compare it with known distributions.

Solution.

Step 1. Write the event as a union of disjoint events.

The variables take only the values $0, 1, 2, \dots$. Thus $x_1 + x_2 = k$ if and only if $x_1 = j$ and $x_2 = k - j$ for one $j \in \{0, \dots, k\}$:

$$\{x_1 + x_2 = k\} = \bigcup_{j=0}^k \{x_1 = j, x_2 = k - j\}.$$

The events are disjoint, because each event has a different value of x_1 .

Step 2. Sum the probabilities of the disjoint events.

$$p(x_1 + x_2 = k) = \sum_{j=0}^k p(x_1 = j, x_2 = k - j).$$

Step 3. Use the independence of x_1 and x_2 .

$$p(x_1 + x_2 = k) = \sum_{j=0}^k p(x_1 = j) p(x_2 = k - j).$$

Step 4. Substitute the Poisson probabilities.

$$p(x_1 + x_2 = k) = \sum_{j=0}^k e^{-\lambda_1} \frac{\lambda_1^j}{j!} \cdot e^{-\lambda_2} \frac{\lambda_2^{k-j}}{(k-j)!}.$$

Step 5. Move the factors that do not depend on j out of the sum.

$$p(x_1 + x_2 = k) = e^{-\lambda_1} e^{-\lambda_2} \sum_{j=0}^k \frac{\lambda_1^j \lambda_2^{k-j}}{j! (k-j)!} = e^{-(\lambda_1 + \lambda_2)} \sum_{j=0}^k \frac{\lambda_1^j \lambda_2^{k-j}}{j! (k-j)!}.$$

Step 6. Multiply and divide by $k!$.

$$p(x_1 + x_2 = k) = \frac{e^{-(\lambda_1 + \lambda_2)}}{k!} \sum_{j=0}^k \frac{k!}{j! (k-j)!} \lambda_1^j \lambda_2^{k-j} = \frac{e^{-(\lambda_1 + \lambda_2)}}{k!} \sum_{j=0}^k \binom{k}{j} \lambda_1^j \lambda_2^{k-j}.$$

Step 7. Use the binomial theorem.

The binomial theorem gives $\sum_{j=0}^k \binom{k}{j} \lambda_1^j \lambda_2^{k-j} = (\lambda_1 + \lambda_2)^k$. Thus

$$p(x_1 + x_2 = k) = e^{-(\lambda_1 + \lambda_2)} \frac{(\lambda_1 + \lambda_2)^k}{k!}.$$

This is the probability mass function of $\text{Poiss}(\lambda_1 + \lambda_2)$.

Conclusion. The sum of two independent Poisson random variables is a Poisson random variable, and the parameters add. By induction, the same is true for any finite number of independent Poisson random variables.

4. Let $x_1, x_2, \dots, x_N$ be an independent sample from $\text{Poiss}(\lambda)$. Find the maximum likelihood estimate of λ .

Idea. Use the standard method for a maximum likelihood estimate:

- Write the likelihood: the probability of the sample as a function of the parameter. For an independent sample, it is a product: $L(\lambda) = \prod_i p(x_i | \lambda)$.
- Take the logarithm: $\ell(\lambda) = \log L(\lambda) = \sum_i \log p(x_i | \lambda)$. The logarithm is an increasing function, so the maximum point does not change.
- Set the derivative to zero, $\ell'(\lambda) = 0$, and solve for the parameter.
- Make sure that the solution is a maximum, for example with the sign of the second derivative $\ell''(\lambda)$.

Solution. Let $S = \sum_{i=1}^N x_i$. Then the sample mean is $\bar{x} = S/N$.

Step 1. Write the likelihood.

$$L(\lambda) = \prod_{i=1}^N e^{-\lambda} \frac{\lambda^{x_i}}{x_i!}.$$

Step 2. Take the logarithm.

The logarithm of a product is the sum of the logarithms:

$$\ell(\lambda) = \sum_{i=1}^N \log \left(e^{-\lambda} \frac{\lambda^{x_i}}{x_i!} \right) = \sum_{i=1}^N (-\lambda + x_i \log \lambda - \log x_i!).$$

Step 3. Calculate each sum.

$$\ell(\lambda) = -N\lambda + S \log \lambda - \sum_{i=1}^N \log x_i!.$$

Step 4. Calculate the derivative.

The last term does not depend on λ :

$$\ell'(\lambda) = -N + \frac{S}{\lambda}.$$

Step 5. Set the derivative to zero and solve.

$$-N + \frac{S}{\lambda} = 0 \implies \frac{S}{\lambda} = N \implies \lambda = \frac{S}{N} = \bar{x}.$$

Step 6. Make sure that the solution is a maximum.

Calculate the second derivative:

$$\ell''(\lambda) = -\frac{S}{\lambda^2}.$$

If $S > 0$, then $\ell''(\lambda) < 0$ for all $\lambda > 0$. Thus ℓ is strictly concave, and the solution of Step 5 is the global maximum.

If $S = 0$, then all $x_i = 0$, and $\ell(\lambda) = -N\lambda$. This function decreases, so the maximum is at $\lambda = 0$. This value is also equal to $\bar{x}$.

The maximum likelihood estimate is the sample mean: $\hat{\lambda}_{\text{ML}} = \bar{x}$.

5. Let $x_1, x_2, \dots, x_N$ be an independent sample from $\text{Poiss}(\lambda)$. The prior distribution is the gamma distribution

$$p(\lambda) = \mathcal{G}(\lambda \mid a, b) = \frac{b^a}{\Gamma(a)} \lambda^{a-1} \exp(-b\lambda), \quad a, b > 0.$$

Find the Bayesian estimate of λ as the mean of the posterior distribution $p(\lambda \mid x_1, \dots, x_N)$.

Idea. Bayes' theorem gives the posterior distribution:

$$p(\lambda \mid X) = \frac{p(X \mid \lambda) p(\lambda)}{\int p(X \mid \lambda) p(\lambda) d\lambda}, \quad X = (x_1, \dots, x_N).$$

The numerator and the denominator contain the same function. The denominator only integrates it over λ . Thus you do not have to calculate the integral directly:

- (a) Write the numerator as $C \cdot g(\lambda)$. The constant C does not depend on λ .

- (b) Find a known distribution q with a density proportional to g : $g(\lambda) = Z q(\lambda)$, where Z is a known constant.
- (c) A density integrates to 1. Thus the denominator is

$$\int CZ q(\lambda) d\lambda = CZ \int q(\lambda) d\lambda = CZ.$$

- (d) Divide the numerator by the denominator:

$$p(\lambda | X) = \frac{CZ q(\lambda)}{CZ} = q(\lambda).$$

The same method gives the posterior mean without a direct integral.

Solution. Let $S = \sum_{i=1}^N x_i$.

Step 1. Write the likelihood as one product.

Collect the factors with $e^{-\lambda}$ and the powers of λ :

$$p(X | \lambda) = \prod_{i=1}^N e^{-\lambda} \frac{\lambda^{x_i}}{x_i!} = \frac{e^{-N\lambda} \lambda^S}{\prod_i x_i!}.$$

Step 2. Multiply the likelihood by the prior.

$$p(X | \lambda) p(\lambda) = \frac{e^{-N\lambda} \lambda^S}{\prod_i x_i!} \cdot \frac{b^a}{\Gamma(a)} \lambda^{a-1} e^{-b\lambda}.$$

Step 3. Separate the constant and the function of λ .

$$p(X | \lambda) p(\lambda) = \underbrace{\frac{b^a}{\Gamma(a) \prod_i x_i!}}_C \cdot \underbrace{\lambda^{a+S-1} e^{-(b+N)\lambda}}_{g(\lambda)}.$$

Step 4. Find the distribution with a density proportional to g .

Let $a' = a + S$ and $b' = b + N$. Then $g(\lambda) = \lambda^{a'-1} e^{-b'\lambda}$. The gamma density with these parameters is

$$\mathcal{G}(\lambda | a', b') = \frac{b'^{a'}}{\Gamma(a')} \lambda^{a'-1} e^{-b'\lambda} = \frac{b'^{a'}}{\Gamma(a')} g(\lambda).$$

Thus

$$g(\lambda) = \frac{\Gamma(a')}{b'^{a'}} \mathcal{G}(\lambda | a', b').$$

Step 5. Calculate the denominator.

Use Step 4. Then use the fact that the gamma density integrates to 1:

$$\int Cg(\lambda) d\lambda = C \frac{\Gamma(a')}{b'^{a'}} \int \mathcal{G}(\lambda | a', b') d\lambda = C \frac{\Gamma(a')}{b'^{a'}}.$$

Step 6. Divide the numerator by the denominator.

$$p(\lambda | X) = \frac{C \frac{\Gamma(a')}{b'^{a'}} \mathcal{G}(\lambda | a', b')}{C \frac{\Gamma(a')}{b'^{a'}}} = \mathcal{G}(\lambda | a', b') = \mathcal{G}(\lambda | a + S, b + N).$$

Step 7. Write the posterior mean as an integral.

Substitute the gamma density:

$$\mathbb{E}[\lambda | X] = \int \lambda \mathcal{G}(\lambda | a', b') d\lambda = \frac{b'^{a'}}{\Gamma(a')} \int \lambda^{a'} e^{-b'\lambda} d\lambda.$$

Step 8. Calculate the integral with the same method.

The function under the integral is proportional to a gamma density with the parameters $a' + 1$ and b' :

$$\lambda^{a'} e^{-b'\lambda} = \lambda^{(a'+1)-1} e^{-b'\lambda} = \frac{\Gamma(a' + 1)}{b'^{a'+1}} \mathcal{G}(\lambda | a' + 1, b').$$

The gamma density integrates to 1. Thus

$$\int \lambda^{a'} e^{-b'\lambda} d\lambda = \frac{\Gamma(a' + 1)}{b'^{a'+1}}.$$

Step 9. Simplify the posterior mean.

Substitute Step 8 into Step 7. Then use $\Gamma(a' + 1) = a' \Gamma(a')$:

$$\mathbb{E}[\lambda | X] = \frac{b'^{a'}}{\Gamma(a')} \cdot \frac{\Gamma(a' + 1)}{b'^{a'+1}} = \frac{\Gamma(a' + 1)}{\Gamma(a') b'} = \frac{a' \Gamma(a')}{\Gamma(a') b'} = \frac{a'}{b'}.$$

The Bayesian estimate is

$$\hat{\lambda}_B = \frac{a'}{b'} = \frac{a + S}{b + N}.$$

Conclusion.

- The gamma prior and the Poisson likelihood give a gamma posterior. A prior with this property is *conjugate* to the likelihood. With a conjugate prior, the update only changes the parameters: $a \rightarrow a + S$ and $b \rightarrow b + N$. Thus a acts as a prior number of events, and b acts as a prior number of observations.
- The estimate is a weighted mean of the prior mean a/b and the maximum likelihood estimate $\bar{x} = S/N$ from Problem 4:

$$\frac{a + S}{b + N} = \frac{a}{b + N} + \frac{S}{b + N} = \frac{b}{b + N} \cdot \frac{a}{b} + \frac{N}{b + N} \cdot \frac{S}{N}.$$

When N increases, the weight $N/(b + N)$ of the data goes to 1, and the estimate goes to $\bar{x}$.

- When $a \rightarrow 0$ and $b \rightarrow 0$, the prior density is proportional to $\lambda^{a-1} e^{-b\lambda} \rightarrow 1/\lambda$. This is the non-informative prior from the notes below. With this prior, the Bayesian estimate is $(0 + S)/(0 + N) = \bar{x}$.

Notes on comparing the estimates from Problems 4 and 5.

1. For more about non-informative priors, see Section 5.4 “Priors” in Kevin P. Murphy, *Machine Learning: A Probabilistic Perspective*.
2. The prior $p(\lambda) \propto \frac{1}{\lambda}$ is a uniform prior on $\log \lambda$. To see this, use the change-of-variables rule for the density of a function of a random variable. See, for example, [this article](#).